

Piazza API

Help for marking the project

To make it easy to review the project, I follow conventional commits (Conventional Commits, no date) and try to keep the commits atomic (What are Git version control best practices?, no date). I create Pull Request for each part of the assessment, where I also try to follow the imperative verb forms, as suggested by the conventional commits project. The Pull requests can be reviewed under the following link: <https://github.com/Meier-Stefan/piazza-api/pulls?q=is%3Apr+is%3Aclosed>

I merge all the branches, but don't delete them, in case it is helpful for review. There are only small code changes in later stages, however, the spelling in the readme is only corrected as of phase G.

Apart from this section, the testing section and the bibliography, this document is a generated Word version of the README.md file.

The following table contains the spell checked parts of the PR messages that aim to comment on how the criteria are met:

Phase A) Install all the necessary packages in the virtual machine: This is done by installing docker and deploying an image. The image contains all the necessary packages because in the Dockerfile on line 6 the command is that stall yarn installs all. The base image is node:lts-alpine, so yarn and node are and installed already. b) Deploy the code: The GitHub Repository is used for de- continuous integration to build the image in the docker cloud. Then the ploy virtual machine pulls the image from there. To make sure that points a) soft- and b) are fulfilled, I checked with the prof first whether it is ok to use ware docker here. c) The setup is described in detail in the report. d) in Discussion of installation and the folder structure: The folder structure vir- that I intend to use is discussed with an example graphic. The models tu- folder does not exist at the moment, because the app does not yet connect alised to the database. To make it simple, in the explanation and the graphic, en- the models are included.

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ments

Phase a) Add user management and JWT functionality using NodeJS. b) Authenticate users for all except the home page. c) Complete verification process for validation purposes.

forcing authentication/verification functionality ties

Phase a) Implement basic functionalities provided in the Action points of Section C: 2. · Authorised users access the Piazza API using JSON web tokens which are used in the OAuth v2 protocol to perform any interaction. · Authorised users post a message for a particular topic in the Piazza API. · Registered users browse messages per topic using the Piazza API. · Registered users perform basic operations, including of “like”, “dislike”, or “comment” a message posted for a topic. · Authorised users can browse for the most active post per topic with the highest likes and dislikes. · Authorised users can browse the REST history data of expired posts per topic. b) Complete requirements as discussed in Phase C section. · Each post on the Piazza wall includes API the following data: · The post identifier. · The title of a post. · The topic of a post from one of the following four categories: Politics, Health, Sport or Tech. Each post could belong to one or more topics. · A timestamp of the post-registration in the Piazza API. · The message body of a post. · The post-expiration time. After the end of this time, the message will remain on the Piazza wall, but it will not accept any further actions, e.g. (likes, dislikes, or comments). · The status of the post, “active” is either true or false · Information about the post owner, id and name · Arrays with ids of the users who liked/disliked the post · Each user interaction with a post on a topic includes the following data: · Information about the user interacting with the post of a topic (id). · The interaction value (including a like, a dislike, or a comment made). · The time left for a post to expire. (This is checked before the action is performed) · The endpoints are listed and explained in the README.md · Only authorised users can access the API. · The API only allows registered users to perform posts.

Phase A) Demonstrate testing of application by either using screenshots of
D: postman, or by developing a testing application e.g., using Node.js or
Test-Python to test endpoints, or by any other use of testing libraries. b)
ing Discuss and document test cases either by including screenshots or code
your samples. c) Complete implementation of the twenty use cases as presented
ap- in section Phase D.

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Phase E) Upload code to a GitHub repository, GitHub Actions -> DockerHub ->
E: Google Cloud VM. An additional command is provided in the report, the
De- rest is already explained in Phase A.

ploy
your
Pi-
azza
project
into
a

VM
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ing
Docker

Phase F) Create a Kubernetes cluster on the Google Cloud platform and deploy
F: five replicas of the app b) Complete requirements for load balancing and
De- replication as discussed in section Phase F

ploy
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in
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netes

a) Provide details on the implementations, database design, service
G: descriptions, API resources, and any other information required. b)
Re- Provide references using the Harvard referencing system as needed in the
port Word and PDF version of the report

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Phase Write the code base highly readable and easy to follow. · Use

H: REST concepts well, as learned in the lecture. · Structure the project

Sub- well to demonstrate understanding of computational complexity in software

mit development. · Don't add comments directly in code, as it is bad

qual-practice. Rather choose meaningful variable names and break complicated

ity transformation into small pieces for better readability. (Special: There is

scripts an additional readme in the filter and sort section, as it gets hard to

follow.) · Implement proper database models, that fit the purpose and

scale of this project. (Meaningful, but not over optimised) · Use

formatter to ensure proper indentation and clear

notations. · Implement simplicity in the flow for

readability. · Add comments in the readme about code pieces that are

very close to what we have seen in class to demonstrate that they are

understood and not simply copy-pasted.

About

A RESTful social media SaaS called Piazza. In Piazza, users post messages for a particular topic while others browse posts and perform basic interactions, including liking, disliking, or adding a comment. This is a coursework project for the Cloud Computing module at Birkbeck, University of London.

Getting started

Install node and yarn or npm on your computer. Create a .env file and populate it with the database connector string that you can retrieve from mongodb.

Setup

Development Setup

The project uses **yarn** as package manager, **prettier** as formatter, **eslint** as linter and **nodemon** to run the development server and track changes during development. These are all listed as 'devdependencies' in the package.json file. I am used to have these helpers in my projects and combined with the VSCode settings, they help me to focus on the logic of the code.

Project Setup

The Api is a Node.js application that uses the Express framework. The docker container that will be deployed to the virtual server in the cloud will have Node installed. This means Node will run the app on the virtual server. During development, however, Nodemon is used, hence it is a development dependency. I set the the eslint config on purpose so that I can use modules, as we use a modern node version and all the dependencies used in the project are compatible.

Phase A

App Directory Structure

All the logic is in the **App** directory. This helps to create a more maintainable and scalable structure for the app. As suggested in this article (Folder Structure for NodeJS & ExpressJS project, 2022) the directory is called **App** and instead of **Src** because the files run on the server and will not be compiled, transpiled, and minified to be sent to the client. The routes forward the request to their corresponding controllers, which will then send the response. This is a good practice shown in the Mozilla Express Tutorial (Express Tutorial Part 4: Routes and controllers - Learn web development | MDN, 2024) Adding the **App** directory as 'imports' alias in the **package.json**, makes it easy to import functions from the right place.

app directory structure

Infrastructure Setup Description

The app is containerized as described in the docker docs the docker docs (Part 1: Containerize an application, 100AD). The only thing that needs to be adjusted is on line 7: the app.js sits in the root directory and is not called index.js.

Docker Build Cloud is used in CI (continuous integration) (Continuous integration, 100AD). For this, the docker username must be added as environment variable in the GitHub Repository. And a Personal Access Token (PAT) needs to be added as secret. GitHub Actions secrets and variables:

Screenshot of the GitHub Actions secrets and variables page

The docker manual triggers on push to the main branch, but I decided to use the manual workflow_dispatch event trigger instead. In the docker build cloud,

a cloud builder needs to be added, I added one and called it **piazza-builder**.

GitHub Actions let the docker cloud build the image:

GitHub Actions let the docker cloud build the image

Docker Build Cloud is fast, but it costs to build. They gave me 60 free minutes because I implemented CI from Github Actions, thanks Docker. Build minutes overview:

Screenshot of build minutes overview

As a result, the image was available on Docker Hub: Docker Hub Images

To make the REST API endpoints available in the virtual machine, Docker was installed on the virtual machine and the docker-user was created like in lab 5.1 (cc/Class-5/Lab5.1 Introduction to Docker.md at master · warestack/cc, no date). Video of lab 5.1 (A Docker tutorial, 2022). The docker-user was logged in on docker hub using the PAT method from above. Then the docker container was started with port mapping like shown in lab 5.1. Because the image was only in the docker cloud, it was automatically pulled before the container was started:

Screenshot of running the container in the virtual machine

Once the container was up and running, the endpoints were available under the virtual machine IP address. The following screenshots show the Home page, post list endpoint, topic detail, topic list and user profile endpoints:

Home page available under the virtual machine IP address

Post endpoint available under the virtual machine IP address

Topic detail endpoint available under the virtual machine IP address

Topic list endpoint available under the virtual machine IP address

User profile endpoint available under the virtual machine IP address

Phase B

Connecting to the database

To start the development server, the node environment is specified in the ‘dev’ script in the package.json file. So the app knows from which .env dotfile to take the environment variables. The main function currently only connects to the database using the **connect** function from mongoose and console logs when it started and if it worked.

Register a user

When a user posts to the **user/register** endpoint, route will forward to the corresponding controller. That controller uses a validation function to make sure that the input makes sense and is secure. The validation function is built with the **string**, **required**, **min**, **max** and **email** functions from the **joi** library.

IDEA: For this purpose, the functions were used in a simple way, but the joi api reference (joi.dev, no date) documents ways how this could be improved.

Using the `bcryptjs` library, the password is hashed and 5 rounds of salt are used before it gets stored in the database.

Login as user

When a user posts to the `user/login` endpoint, there is, additional to the validation with `joi`, the password gets compared with a function from `bcryptjs` and, if the password matches, the response contains a json web token, that can later be used to access resources that require authentication.

The `jsonwebtoken` library creates that token using a secret variable from the `.env` dotfile.

SPECIAL: The json web token expires after one hour and contains the user ID as payload (jsonwebtoken, 2023). This user Id can for example be used by the controllers to make sure that one user does not modify the profile of another user.

IDEA: It would be useful if the token would refresh.

Interact with an endpoint that requires authentication

To keep the repository clean and organized, a middleware folder is used like suggested by this tutorial (Node.js Authentication and Authorization with JWT: Building a Secure Web Application, 2023). The routes that require use the `isAuthenticated` middleware function to make sure that the user is logged in and allowed to proceed. Logged in means that the request headers contain a key value pair of a key containing `auth-token` and a value containing the json web token. If the user is not logged in, or has a wrong json web token, `isAuthenticated` will send the corresponding response instead of letting the route forwarding the request to the controller.

Phase C

Models

The Models are created so that they are practical to use in this small project.

As Joe Karlsson from mongodb explains in this video (MongoDB Schema Design Best Practices, 2021), it is a good idea to think about what will be displayed when designing the database schema. I tried to follow this advice, but also, I did try not to over optimise as if this app was serving millions of users already.

There are 3 Models:

UML diagram about models

The above diagram follows the style of the Mozilla tutorial (Express Tutorial Part 3: Using a Database (with Mongoose) - Learn web development | MDN, 2024) that also uses mongoose. The maximum and minimum of each model that may be present in the relationship are at the corresponding end of the connection

line. An asterisk (*) means that there is theoretically no maximum. Practically, as mentioned above, my implementation can reach the 16MB document limit of MongoDB.

Users can create zero or many **Posts**, **Comments**, **Likes**, and **Dislikes**. Each **Post** and each **Comment** have one **User** as author. **Likes** and **Dislikes** are arrays that can have zero or many **Users** listed. Each **Post** must have at least one **Topic**, but a **Topic** has zero or many **Posts**.

Post

Screenshot of the DB entry

The comments are and array of documents in the post documents for now. This is how it will be displayed.

Reactions and comments contain the link to the user like this: `likes: [{ type: Schema.Types.ObjectId, ref: "users" }],`

IDEA: For projects that could have millions of users, it would make sense to store the comments in separate collections and also to track likes in documents that store which user liked/disliked which post. Both of the above will help to avoid problems with the 16mb document limit and improve the query speed.

Topic

Screenshot of the DB entry

The topic is simple, it just contains a title and an array of posts. However, currently I don't use that array. If I display the posts of a topic, I use the post list controller and filter for the topic id.

IDEA: The likes and dislikes are arrays of `userIds`. This can be displayed as number in the frontend. The user can then click/hover on these reaction numbers to see who reacted. Also, the creation date could be made human readable like the time to expiration in the `showIfActive` function.

User

Screenshot of the DB entry

The username is not unique, so there could be more than one user with the same name, but for linking, the user ID is stored. Like this, buttons like `reply in direct message` or so, will still work.

Reactions and comments

The reactions and comments on posts can only be done from the post detail endpoint. The frontend can call that from the list view and only update the affected post though.

Filter and Sort

With this solution, the database stores the arrays and if the user wants to sort by interest, the app has to count and sort.

Similarly the status 'active' is calculated by the app with the help of database values and the local time of the client.

Because both of the above modify the posts that are displayed and can be filtered or sorted, the logic gets a bit messy. One sort or filter is done locally and based on that, the rest is sent to the database to handle.

IDEA: The limit of posts fetched is set to 15, which is a good number of posts to show per page. The frontend pagination logic will trigger fetching the next 15 if the user paginates.

API reference

/:

The welcome page. Responds with 'please log in' or if logged in, 'hello, [username]'.

Endpoints that can be accessed if authorised:

GET post/:

List of posts. Responds with an array of Post objects. Can be used with **filters** or **sort**, that are passed in the request body as follows: `{filters:[options]}` options include:

active: type: Boolean `false|true`

topic: type: Schema.Types.ObjectId `"topicId"`

authorId: type: Schema.Types.ObjectId `"topicId"`

POST post/new:

Create a new post. Responds with the newly created post object. The new post is passed in the request body in the following format:

```
title: { type: String, required: true, min: 3, max: 256 },
text: { type: String, required: true, min: 3, max: 2048 },
topicId: [{ type: Schema.Types.ObjectId, required: true, ref: "topics" }],
expirationTime: { type: Number, required: true, min: 5 },
```

example:

```
{"title":"Question", "text":"What is the coolest animal?",
"expirationTime":"8", "topicId": "674c7965f3a45b229d3b02d9"}
```

GET post/:id:

Detail view of a post.

POST post/:id/comment:

Accepts new comments for posts if they are still active. The comments are passed in the request body in the following format:

```
{ text: { type: String, required: true, min: 3, max: 1024 }}
```

POST `post/:id/react`:

Accepts new reactions for posts if they are still active. Reactions are passed in the request body in the following format:

```
{ reaction: [options]}
```

The options are one of the two following strings: "likes" or "dislikes"

GET `topic/`:

List of all the topics. Still needs to be implemented. Currently responds with the string "Topic list endpoint"

GET `topic/:id`:

Detail view of a topic. Responds with a list of post that belong to the topic. Accepts the same request body options as `post/`.

POST `user/login`:

Login page. Responds with a json web token. This is implented in that way that it is easy to test/review. Accepts the following request body:

```
{email: { type: String, required: true, min: 3, max: 256 },  
password: { type: String, required: true, min: 3, max: 1024 },}
```

example:

```
{"email": "user@mail.com",  
"password": "TQVm4cBK"}
```

POST `user/registration`: Registration page. Responds with the new created user object. Accepts the following request body:

```
username: { type: String, required: true, min: 3, max: 256 },  
email: { type: String, required: true, min: 3, max: 256 },  
password: { type: String, required: true, min: 3, max: 1024 }
```

example:

```
{"username": "user",  
"email": "user@mail.com",  
"password": "TQVm4cBK"}
```

GET `user/:id`

User detail page. Still needs to be implemented. Currently responds with the string "User profile endpoint".

Phase D

To test the application, the app was deployed in the google cloud virtual machine and then tested manually.

TC1:
Olga, Nick, Mary, and Nestor register and are ready to access the Piazza API.
TC1
TC1
TC1
TC1
TC1
TC1
TC1
TC1
TC2:
Olga, Nick, Mary, and Nestor use the oAuth v2 authorisation service to register and get their tokens.
TC2
TC2
TC2
TC2
TC2
TC2
TC2
TC2
TC3:
Olga makes a call to the API without using her token. This call should be unsuccessful as the user is unauthorised.
TC3
TC4:
Olga posts a message in the Tech topic with an expiration time (e.g. 5 minutes) using her token. After the end of the expiration time, the message will not accept any further user interactions (likes, dislikes, or comments).
TC4
TC5:
Nick posts a message in the Tech topic with an expiration time using his token.
TC5
TC6:
Mary posts a message in the Tech topic with an expiration time using her token.
TC6
TC7:
Nick and Olga browse all the available posts in the Tech topic; three posts should have zero likes, zero dislikes, and no comments.
TC7
TC7
TC8:
Nick and Olga “like” Mary’s post on the Tech topic.
TC8
TC8
TC9:

Nestor “likes” Nick’s post and “dislikes” Mary’s on the Tech topic.

TC9

TC9

TC10:

Nick browses all the available posts on the Tech topic; at this stage, he can see the number of likes and dislikes for each post (Mary has two likes and one dislike, and Nick has one like). No comments have been made yet.

TC10

```
[
  {
    "_id": "6755f4ad9502ff8e22d4ef69",
    "title": "Best invention",
    "text": "What is the best technology invented? Leave it in the comments below.",
    "authorId": "6755f2cf9502ff8e22d4ef5f",
    "authorName": "Mary",
    "topic": [
      "6755f202b01558af18266944"
    ],
    "expirationTime": 50,
    "likes": [
      "6755f2969502ff8e22d4ef59",
      "6755f2ba9502ff8e22d4ef5c"
    ],
    "dislikes": [
      "6755f2e29502ff8e22d4ef62"
    ],
    "comments": [],
    "date": "2024-12-08T19:34:05.048Z",
    "__v": 0,
    "active": true,
    "timeLeft": "24 minutes"
  },
  {
    "_id": "6755f5319502ff8e22d4ef6d",
    "title": "I know the best invention",
    "text": "I truly believe that Windows XP was the peak of tech inventions.",
    "authorId": "6755f2ba9502ff8e22d4ef5c",
    "authorName": "Nick",
    "topic": [
      "6755f202b01558af18266944"
    ],
    "expirationTime": 50,
    "likes": [
      "6755f2e29502ff8e22d4ef62"
    ],
  },
]
```

```

        "dislikes": [],
        "comments": [],
        "date": "2024-12-08T19:36:17.375Z",
        "__v": 0,
        "active": true,
        "timeLeft": "26 minutes"
    },
    {
        "_id": "6755f5e89502ff8e22d4ef71",
        "title": "Check out that Channel on Youtube",
        "text": "Techworld with Nana: https://www.youtube.com/@TechWorldwithNana",
        "authorId": "6755f2969502ff8e22d4ef59",
        "authorName": "Olga",
        "topic": [
            "6755f202b01558af18266944"
        ],
        "expirationTime": 5,
        "likes": [],
        "dislikes": [],
        "comments": [],
        "date": "2024-12-08T19:39:20.568Z",
        "__v": 0,
        "active": false
    }
]

TC11:
Mary likes her post on the Tech topic. This call should be unsuccessful; in
Piazza, a post owner cannot like their messages.
TC11
TC12:
Nick and Olga comment on Mary's post on the Tech topic in a round-robin
fashion (one after the other, adding at least two comments each).
TC12
TC12
TC12
TC12
TC13:
Nick browses all the available posts in the Tech topic; at this stage, he can see
the number of likes and dislikes of each post and the comments made.
TC13

[
    {
        "_id": "6755f4ad9502ff8e22d4ef69",
        "title": "Best invention",
        "text": "What is the best technology invented? Leave it in the comments below.",

```

```

    "authorId": "6755f2cf9502ff8e22d4ef5f",
    "authorName": "Mary",
    "topic": [
        "6755f202b01558af18266944"
    ],
    "expirationTime": 50,
    "likes": [
        "6755f2969502ff8e22d4ef59",
        "6755f2ba9502ff8e22d4ef5c"
    ],
    "dislikes": [
        "6755f2e29502ff8e22d4ef62"
    ],
    "comments": [
        {
            "commentorId": "6755f2ba9502ff8e22d4ef5c",
            "text": "I know it, and by coincidence posted about it... It is Windows XP!",
            "_id": "6755fe28a176b8e27d6cdca9",
            "date": "2024-12-08T20:14:32.467Z"
        },
        {
            "commentorId": "6755f2969502ff8e22d4ef59",
            "text": "I believe you are wrong, Nick! It is cloud computing!",
            "_id": "6755fe92a176b8e27d6cdcaf",
            "date": "2024-12-08T20:16:18.575Z"
        },
        {
            "commentorId": "6755f2ba9502ff8e22d4ef5c",
            "text": "Nono! I believe you are wrong, Olga! There was even a cloud on the",
            "_id": "6755fefda176b8e27d6cdcb7",
            "date": "2024-12-08T20:18:05.584Z"
        },
        {
            "commentorId": "6755f2969502ff8e22d4ef59",
            "text": "Nick, clouds are mostly made out of linux computers...",
            "_id": "6755ff3fa176b8e27d6cdcc1",
            "date": "2024-12-08T20:19:11.111Z"
        }
    ],
    "date": "2024-12-08T19:34:05.048Z",
    "__v": 0,
    "active": true,
    "timeLeft": "a minute",
    "engagement": 3
},
{

```

```

    "_id": "6755f5319502ff8e22d4ef6d",
    "title": "I know the best invention",
    "text": "I truly believe that Windows XP was the peak of tech inventions.",
    "authorId": "6755f2ba9502ff8e22d4ef5c",
    "authorName": "Nick",
    "topic": [
        "6755f202b01558af18266944"
    ],
    "expirationTime": 50,
    "likes": [
        "6755f2e29502ff8e22d4ef62"
    ],
    "dislikes": [],
    "comments": [],
    "date": "2024-12-08T19:36:17.375Z",
    "__v": 0,
    "active": true,
    "timeLeft": "3 minutes",
    "engagement": 1
  },
  {
    "_id": "6755f5e89502ff8e22d4ef71",
    "title": "Check out that Channel on Youtube",
    "text": "Techworld with Nana: https://www.youtube.com/@TechWorldwithNana",
    "authorId": "6755f2969502ff8e22d4ef59",
    "authorName": "Olga",
    "topic": [
        "6755f202b01558af18266944"
    ],
    "expirationTime": 5,
    "likes": [],
    "dislikes": [],
    "comments": [],
    "date": "2024-12-08T19:39:20.568Z",
    "__v": 0,
    "active": false,
    "engagement": 0
  }
]

```

TC14:

Nestor posts a message on the Health topic with an expiration time using her token.

TC14

TC15:

Mary browses all the available posts on the Health topic; at this stage, she can

see only Nestor's post.

TC15

TC16:

Mary posts a comment in Nestor's message on the Health topic.

TC16

TC17:

Mary dislikes Nestor's message on the Health topic after the end of post-expiration time. This should fail.

TC17

TC18:

Nestor browses all the messages on the Health topic. There should be only one post (his own) with one comment (Mary's).

TC18

TC19:

Nick browses all the expired messages on the Sports topic. These should be empty.

TC19

TC20:

Nestor queries for an active post with the highest interest (maximum number of likes and dislikes) in the Tech topic. This should be Mary's post.

TC20

TC20

Phase E

For the testing, the app was deployed into a google cloud virtual machine with almost the same method as described above in Phase A: From my local, I push code to Github, and there I manually trigger a Github Action, that lets the docker cloud build the image. After that, the image is pulled from the cloud VM, that has Docker installed and a docker user set up. The only difference to Phase A is, that now there is an app deployed that needs to handle tokens and connect to the database. As I am not pushing the .env file to github, the secrets must be provided differently: To provide the env variables needed, I passed them to the Docker container using the `--env` flag: `docker run --env MONGODB_URI"<the actual string>" --env JWT_SECRET="<the actual secret>" -p 80:3000 meierstefan/piazza-api` There are better ways, but for this one time deployment it is fine like this.

Phase F

5 Replicas

In kubernetes, secrets can not simply be passed as env variables, because there are multiple containers and maybe we will add more later. To provide the secret in the simplest way, I created Kubernetes secrets as follows:

```
kubect1 create secret generic database-connection \
```



```
--from-literal=MONGODB_URI='<actual string that the mongodb dashboard provides>'
```

```
kubectl create secret generic json-web-token-secret \
--from-literal=JWT_SECRET='<actual secret>'
```

Then everything was ready and the fife nodes could be started like we learned in class (cc/Class-6 at master · warestack/cc, no date, p. 6). First the deployment yaml file is created with vim directly in the google cloud console.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: piazza-api-deployment
  labels:
    app: piazza-api
spec:
  replicas: 5
  selector:
    matchLabels:
      app: piazza-api
  template:
    metadata:
      labels:
        app: piazza-api
    spec:
      containers:
        - name: piazza-api
          image: meierstefan/piazza-api:latest
          imagePullPolicy: Always
          ports:
            - containerPort: 3000
          envFrom:
            - secretRef:
                name: json-web-token-secret
            - secretRef:
                name: database-connection
```

The simple steps are explained one by one in this article (Kubernetes Deployment YAML File with Examples, no date). Interesting is the `imagePullPolicy`: Here (Service, no date) the kubernetes docs explain what is going on. We have set it so **Always**, which means that everytime a new container is launched, kubernetes checks, if there is a new image in the docker registry. If there is a new one, it pulls it to the virtual machine. But if the one in the cache is the exactly same already, it does not pull the image because it can use the cache.

To start the replicas, we tell `kubectl` to apply the file. Kubernetes start of replicas.

Kubernetes start loadbalancer service.

Figure 1: Kubernetes start loadbalancer service.

To make sure that they are alive, we tell it to get the pods.
Kubernetes get pods.

IDEA: To make the setup easier, it would be a good idea to put the secrets into the secret-manager (Secret Manager, no date) and then they could be accessed from terraform. Then, it is no longer necessary to manually start the pods with `kubect1`.

Loadbalancer

With the load balancer, we make the pods accessible. Like above, we need this file first:

```
apiVersion: v1
kind: Service
metadata:
  name: piazza-api-service
  labels:
    app: piazza-api-service
spec:
  type: LoadBalancer
  ports:
    - name: http
      port: 80
      protocol: TCP
      targetPort: 3000
  selector:
    app: piazza-api
  sessionAffinity: None
```

The docs (Service, no date b) explain that kubernetes does not have a load balancer, but by setting the `type` of this service to `LoadBalancer`, we will use the google cloud default load balancer. We set `sessionAffinity` to `None` because it does not matter if the request of the client is always passed to the same pod.

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